

A disproof of the conjectured recurrence for OEIS A119967, and the recurrence that holds instead

Adrian Perez Fontelles

Independent researcher

30 August 2026

Abstract

OEIS A119967 carries a conjectured linear recurrence with polynomial coefficients, of order 4. It is false. The entry's own generating function gives, exactly, the quantity the recurrence claims to annihilate, and that quantity is not eventually zero: the smallest failure is at $n = 4$, where the left-hand side equals 54 rather than 0, and the failure recurs for infinitely many n . We then give a recurrence of order 6 that does hold, derived from the same generating function and proved by the same criterion that refutes the conjectured one.

2020 Mathematics Subject Classification. 05A15, 11B37, 33F10.

1 The sequence and the conjecture

OEIS A119967 is “A transform of the central binomial coefficients $C(n, \text{floor}(n/2))$ ”. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, 1, 2, 4, 9, 21, 49, 113, \dots$$

The entry states, not as a conjecture,

$$\text{G.f.: } 2*(1-x)/(1-2*x-x^2+\sqrt{1-4*x+6*x^2-4*x^3-3*x^4}).$$

that is,

$$A(x) = \sum_{n \geq 0} a(n)x^n = \frac{2(x-1)}{x^2 + 2x - \sqrt{-3x^4 - 4x^3 + 6x^2 - 4x + 1} - 1}. \quad (1)$$

This is the only input taken from the entry, and it was checked against every term of the entry's DATA section before use — all 31 of them — not against a sample.

The entry separately carries the following comment:

Conjecture: $(n+2)*(n-3)*a(n) + (2+2*n-n^2)*a(n-1) + (3*n-5)*(n-2)*a(n-2) - (n-2)*(n-3)*a(n-3) - (n-2)*(n-3)*a(n-4) = 0$. - *R. J. Mathar*, Nov 14 2011

As of the “Last modified” line on the live entry (2017-03-19, revision 16) the statement is still recorded as a conjecture, and nothing on the entry records it as settled either way.

Written out, the conjecture asserts

$$(n-3)(n+2)a(n) + (-n^2+2n+2)a(n-1) + (n-2)(3n-5)a(n-2) - (n-3)(n-2)a(n-3) - (n-3)(n-2)a(n-4) = 0 \quad (2)$$

2 The residual, and what it decides

Write $\theta = x \frac{d}{dx}$, so that $\theta x^n = n x^n$, and set

$$B(x) = \sum_{i=0}^4 x^i (p_i(\theta + i)A)(x). \quad (3)$$

Lemma 1. $[x^n]B(x) = \sum_i p_i(n) a(n-i)$ for every n . Consequently the recurrence (2) holds for all $n > d$ if and only if B is a polynomial of degree at most d ; and if B is not a polynomial at all, (2) fails for infinitely many n .

Proof. θ acts diagonally on powers, so $(p_i(\theta + i)A)(x) = \sum_m p_i(m+i)a(m)x^m$, and multiplying by x^i shifts the exponent, giving $\sum_m p_i(m+i)a(m)x^{m+i} = \sum_n p_i(n)a(n-i)x^n$. Summing over i gives the first claim. The rest is immediate: a power series is a polynomial of degree at most d exactly when all its coefficients beyond x^d vanish, and a series with infinitely many nonzero coefficients is not a polynomial. $\square$

Lemma 1 is the same criterion used to *prove* such conjectures. Its force here is that it is an equivalence, so a negative answer is a disproof rather than a failure to find a proof — and the answer is computed exactly, in the field $\mathbb{Q}(x)[y]/(P(x, y))$ where A lives, with no series truncated.

3 The conjecture is false

Carrying out (3) for A119967 and reducing in $\mathbb{Q}(x)[y]/(P(x, y))$, the residual B is *not* a polynomial. By Lemma 1 the conjectured recurrence therefore fails for infinitely many n .

The failure is not confined to small indices, and it is not an artefact of the first few terms: expanding (1) and evaluating the left side of (2) directly gives

n	$\sum_i p_i(n)a(n-i)$
4	54
5	279
6	1110
7	3871
8	12438
9	37933
10	111902
11	322692
12	914218
13	2551049

Theorem 2. The recurrence (2) conjectured on OEIS A119967 is false. Its smallest failure is at $n = 4$, where the left-hand side equals 54, and it fails for infinitely many n .

Proof. The tabulated values are computed from (1), whose coefficients agree with the entry's DATA at every published index, so each is a value of $\sum_i p_i(n)a(n-i)$ for the sequence A119967 itself; the entry at $n = 4$ is nonzero. That B is not a polynomial, established exactly above, gives infinitely many such n by Lemma 1. $\square$

Remark 3. The failure is not a matter of how the recurrence is indexed. Shifting the argument of the coefficients and the indices of the terms together, by any amount between -5 and 5 , does not make it hold on the published data either.

4 A recurrence that does hold

The generating function (1) is algebraic over $\mathbb{Q}(x)$, so the sequence is P -recursive and a correct recurrence exists; it is obtained from (1) rather than guessed. Differentiating the algebraic equation satisfied by A expresses A' inside the field $\mathbb{Q}(x)[y]/(P)$, so $A, A', \dots$ span a finite-dimensional space over $\mathbb{Q}(x)$ and satisfy a linear differential equation with polynomial coefficients. Reading off the coefficient of each power of x turns that equation into a recurrence.

Theorem 4. *For all $n \geq 6$,*

$$(-n^2 - 2n) a(n) + (6n^2 + n - 4) a(n-1) + (-15n^2 + 29n - 4) a(n-2) + (20n^2 - 82n + 72) a(n-3) + (-11n^2 + 8n - 12) a(n-4) = 0$$

Proof. The residual (3) formed with these coefficients reduces, in $\mathbb{Q}(x)[y]/(P(x, y))$, to 0 identically. By Lemma 1 the recurrence therefore holds at every n for which it is stated, that is for $n \geq 6$. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the generating function (1) was expanded and compared against the entry's DATA at every one of the 31 published indices; the values agree exactly. A generating function that reproduced only some of them would not have been used.

Second, the conjectured recurrence was evaluated on those published terms in exact integer arithmetic, with no generating function involved: it is nonzero at 10 of the 10 indices tabulated above.

Third, the recurrence of the previous section was evaluated the same way, on the published terms, and vanishes at all 25 indices where it applies.

References

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